

Population Aging and Housing Prices: Evidence from OECD Countries

MinSeok (Michael) Cho

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ABSTRACT

This research paper examines the relationship between the old age dependency ratio and the real house price index, inferring the potential correlation between the aging population and the housing price. We utilized panel data that is composed of 31 OECD countries spanning from 2010 to 2023.

This paper employs several models, from pooled OLS, random effects, to fixed effects. Heteroskedasticity in the error terms is assessed using the Wald Test, and the preferred econometric model is selected based on the Hausman Test, which confirms that fixed effects is the appropriate model over random effects in this research.

After applying entity and year fixed effects and confirming through the Hausman Test that fixed effects is the appropriate, we find out that the old age dependency ratio and the real house price index has a negative relationship – one percentage point increase in the old age dependency ratio is associated with a decrease in 2.316 in the housing price index, holding other variables constant. Absorbing common unobserved time characteristics, there was a constant increase in global housing price appreciation from 2013 and onwards, and showed sharp increase during 2020-2021, coinciding with the COVID era and low interest environment, and a partial decrease from 2022 to 2023.

Based on the findings from the key and additional explanatory variables, we propose the policy pathways that can comprehensively prevent the older generations to fail their retirement plan or lose the value of residential properties, but allow the younger generations to gain a foothold that will make them to gain independent home, and actively participate in the economy at the same time.

INTRODUCTION

The housing price is constantly increasing, and it's decreasing housing affordability. In Korea, the *Korea Real Estate Board's* 2026 January trends reported that Seoul apartment sales prices rose 0.91% from the previous month, and the *Seoul Metropolitan Government* official announced there was an increase of 13.5% year-over-year from December 2024 to December 2025.

Motivated by South Korea's housing market and possible issues in affordability to the younger generations, this research paper observes how the aging population is associated with the housing price by looking at the old age dependency ratio and the real house price index. According to the *Korea Research Institute for Human Settlements*, the average age of first-time purchasing households in the last four years was 46.4 as of 2024, approximately 14.1 years older than in 1990 to 1994.

At the same time, a much lower proportion of the younger generation owns homes, compared to the older generations. The South Korean *Ministry of Data and Statistics* published a press release stating that over 18.7% of those aged 70 or older are homeowners, while 10.6% are aged 40 or younger, and this is likely to accelerate due to an aging population.

These trends are not isolated to South Korea alone. Across OECD countries, demographic shifts driven by declining fertility rates and increasing life expectancy have begun to reshape housing demand patterns and supply patterns. As the share of older populations grows relative to the working-age population, questions arise about whether aging demographics systematically suppress or elevate housing prices, and what the additional driving factors are. Understanding this relationship has direct implications for housing policy, urban planning, and intergenerational wealth distribution.

With the combination of the aging population, what effect or association would be present between the old age dependency ratio and the housing price? From this research project, we would like to assess and address the concerns about the current demographic structure and how this could affect the housing market by looking at the price index. In addition, the variables considered in the housing purchase decision are long-term interest rate, unemployment rate, core CPI, net savings rate, and real GDP.

In this research paper, this paper uses country-year panel data, using single linear regression, multiple linear regression, and fixed and random effects model to compare which model is more suitable for this topic, using Hausman Test. After comparing and analyzing several models, we found out that the old age dependency ratio and the housing price index have a negative relationship, meaning that one percentage point increase in the old age dependency ratio is associated with a decrease in the housing price index.

This paper will review similar studies that have already been conducted, and explain the dataset we will use in this research with visualizations that reflects the trends of the real housing price index, and how relevant factors have changed over the years. Then, major econometric models will be shown, and results that include potential policy implementation.

LITERATURE REVIEW

The relationship between population aging and housing prices has been studied across a range of countries and methodological approaches, with findings that differ by context, time period, and the entity involved in each research.

The most directly comparable study is *Sungwon Cho's* (2021) examination of the causal relationship between population aging and housing prices in South Korea. Using quarterly time-series data from 1990 to 2020, *Cho* employed a Granger causality test within an error correction

model framework, with the old-age dependency ratio (population aged 65 and over divided by the working-age population (ages 15–64)) as the measure of aging, and the Bank of Korea's housing price index as the dependent variable. *Cho* found a unidirectional causality from population aging to housing prices and indicated that population aging can be used for forecasting housing prices, while housing prices did not significantly cause changes in population aging. This implies that past values of the old-age dependency ratio have predictive power over future housing prices in Korea. Our paper is similar to *Cho*'s in that we also use the old-age dependency ratio as the key independent variable and examine its association with a housing price index. A key difference is that our study uses OLS regression rather than a Granger causality framework and focuses on the level association between the two variables rather than their predictive power.

Jinyhup Kim's (2024) study broadens the scope to the United States housing market, examining how population aging affects housing values and rents across various housing types and sizes. This study used panel data consisted of 1820 US counties from 2010 to 2020, and Zillow's Housing Value Index (ZHVI) and Zillow Observed Rent Index (ZORI) for dependent variables, and independent variables such as old-age dependency ratio, young adults' homeownership, logged total population, and more. Using fixed effects models across 1,820 counties, *Kim* found that a higher old-age dependency ratio was associated with increase of 25.1% increase in housing values but associated with decrease of 57.5% in rental prices. This suggests that older adults tend to age in place or downsize within owner-occupied housing rather than transitioning to rentals, sustaining demand for sales housing. Our research shares common factor of using the old-age dependency ratio as the key explanatory variable, though we focus specifically on the real house price index rather than segmenting by housing type or tenure, and our analysis operates at a different geographic scale.

The paper from *Hongbin Li* addresses how demographic changes, such as longevity and old age dependency rate, are associated with savings and investment. *Li* used panel data from the World Bank over 200 countries across years from 1960 to 2004, and ran three different regressions: savings (savings to GDP ratio), investment (investment to GDP ratio), and growth rate (annual growth rate of per capita GDP). Throughout these three dependent variables, *Li* focused on the independent variables such as old (aged 65+) to middle-age (15-64) population, life expectancy, fertility rate, labor force participation rate, and GDP per capita. Looking at the investment regressions, it showed that one percentage point increase in old to middle-age population is associated with decrease in 0.472 percentage point in investment to GDP ratio. This could infer that economy invests a smaller fraction of its output when the society gets older. Though *Li*'s focus is on savings and investment rather than housing prices directly, their findings are relevant with our research because investment and housing prices are closely linked in a way that a declining investment rate with aging society may dampen housing demand and construction activity, with decrease on price levels.

Taken together, these three studies suggest that population aging, measured by the old-age dependency ratio, has measurable and significant associations with economic outcomes, ranging from housing prices, savings and to investments. While findings on the direction of the effect on housing prices are not equivalent across countries and contexts, the weight of evidence points toward aging as a meaningful predictor of housing market dynamics. Our paper contributes to this literature by examining this relationship with a specific focus on the real house price index, aiming to provide a clean and interpretable estimate of how shifts in the old-age dependency ratio correspond to changes in housing prices.

DATA AND DESCRIPTIVE STATISTICS

The majority of the data was exported from the OECD Data Explorer. The original data included all 38 countries, with the time periods available varying by the variables. **Table 1** shows the description of the variables. The main X variable is the old age dependency ratio (*oad_r* in dataset), and the main Y variable is the analytical house price indicators (*house_ind* in dataset). The old age dependency ratio is measured as the number of people aged 65 or older as a percentage of the working-age population aged 20–64. The house price indicators, or real house price index, is the measure of the prices of residential properties over time, with a base year in 2015. The additional independent variables include Core CPI, long-term interest rate, unemployment rate, average annual wages, real GDP per capita, and net savings rate. Core CPI is included to capture how inflation dynamics transmit into housing price growth. As food and energy are volatile and might disrupt the stable analysis, we decided to use Core CPI. Since housing is often considered an inflation-sensitive asset, we expect a positive relationship. A coefficient greater than 1 would indicate that housing price growth amplifies inflationary changes more than one-for-one, suggesting housing markets are particularly reactive to inflation. Long-term interest rate was included to refer the mortgage rate. As mortgage rate tends to follow and has less gap than the short-term interest rate, we instead chose long-term interest rate to indirectly infer the mortgage rate and its trend. Unemployment rate measures the percentage of the total labor force aged 15 and over that is unemployed, and is calendar and seasonally adjusted on an annual basis. We expect this to reflect how the nation's economy is vital in labor force. Average annual wages are measured in constant 2024 USD prices, and shows each country's annual average income. Real GDP per capita is also normalized to constant 2020 USD dollars per person. This would reflect the overall economic status of the nation. The net savings rate is measured as a percentage of GDP, derived by subtracting depreciation from gross savings, and is included to control for the economy's aggregate capital accumulation capacity. We included this variable to capture the broader macroeconomic environment in which housing investment decisions are made in each country. Some variables – real GDP and average salary – are normalized to purchasing power parity (PPP) because all countries have different cost of living, and we believed this would mitigate the economic situations.

As a result, the total observation (entity – year pairs) were 429. We trimmed 38 OECD countries to 31, based on the external data availability. The deleted countries are Colombia, Costa Rica, Estonia, Iceland, Mexico, New Zealand, and Türkiye. These countries are deleted if either 1) we couldn't find any external data, or 2) if more than 10 years of data are missing. With the remaining data, we then dealt with finding and filling in some alternative values, such as Switzerland's unemployment rate. Because Federal Reserve Bank of St. Louis' FRED owned it which had the same definition, we filled in the value of its *unemp_rate* variable. We still had missing variables in Japan's Core CPI (2021-2023), real GDP (2023), Norway's real GDP (2022-2023), and net savings (2023). Though these values were missing, we decided not to exclude these two countries as there were not that many missing values; instead, we removed those years in two countries.

Table 2 shows the descriptive statistics of the variables that are used in this study. The variable *country_id* and *year* is used to run the fixed and random effects for this panel data. The housing price index's lowest record was from Chile in 2010, about 30 indices below the baseline of 2015. Meanwhile, Hungary in 2022 recorded the highest housing price index among the 31 OECD

members. The old age dependency ratio showed lowest across the years from Chile in 2010, while Japan recorded the highest in 2020 among the selected OECD countries.

Plot 1 shows the trend between the average of old age dependency ratio and the real house price index by year. It kept decreasing until 2013 but started to increase until 2022. In the period between 2020 to 2022 showed a big jump in real house price index, potentially from the pandemic and the increase of demand in remote work, which increased the good quality of housing. One another thing to note is that the old age dependency ratio slightly decreased from 2020 to 2021. Countries like Canada mentioned that COVID-19 and opioid crisis influenced a decline in life expectancy,¹ resulting in reduction in both the numerator (population aged 65 or older) and denominator (usually population aged in between 15 to 64).

Table 3 shows the descriptive statistics, comparing the averages of explained and explanatory variables based on Pre- and Post-COVID periods. After running t-test, all variables except the *avg_salary* and *net_save* showed statistically significant at 5% significance level. This means that we fail to reject the null hypothesis that the means are equal for average salary and net savings rate. In other words, there is insufficient statistical evidence that the pre- vs. post-COVID difference in those two variables is meaningful. In addition, as we can see that core CPI is statistically significant while average salary is not, we can infer that the real wages likely fell. Core CPI, or inflation, surged more aggressively than nominal salaries did Post-COVID. In addition, net savings rate having statistically insignificant reflects that households and economies maintained relatively stable saving behavior. Though Pre-COVID has slightly lower mean of net savings rate compared to the entire observations' mean on net savings rate, the difference (0.123) is negligible as it is lower than its entire data's standard deviation (6.042). However, because the dataset we have more years of Pre-COVID (from 2010 to 2019), the equal comparison between the Pre- and Post-COVID value might be cautious.

ECONOMETRIC MODEL

For this research, we propose the below universal econometric model that focuses old age dependency ratio, or *oad_r*, as the key independent variable, and observes how the explanatory variables are associated with the real house price index, or *house_ind*:

$$\text{house_ind} = \beta_0 + \beta_1 \text{oad_r} + \beta_2 \text{avg_salary} + \beta_3 \text{core_cpi} + \beta_4 \text{lt_ir} + \beta_5 \text{net_save} + \beta_6 \text{real_gdp} + \beta_7 \text{unemp_rate} + \alpha_i + \gamma_t + \varepsilon_{it}$$

which represents all the independent variables, regardless of their statistical significance, and the unobserved effects of entity, time, and error term. The α_i represents each country's unobserved characteristics, γ_t represents each year's unobserved characteristics, and ε_{it} represents the error term that varies across countries and time.

Because there are several countries and years in this dataset, and the regression might consider each country and year pair as totally different observation, there is a heterogeneity or unobserved error issue. In order to combat this problem, we will try to utilize several ways. First, utilizing random effects will assume the country specific unobserved effects' differences as uncorrelated with the explanatory variables, and models them as part of the error term instead. From this model, knowing a country's characteristics tells nothing about its unobserved characteristics. Second, utilizing fixed effects will assume country specific unobserved effects' differences are

¹ (Canada)

correlated with the explanatory variables, and it will remove the errors by demeaning, so the explanatory variables can't bias the coefficients.

Given the structural differences across countries in our sample, heteroskedasticity is likely present. Specifically, the variance of residuals is expected to differ systematically across countries, as more volatile housing markets such as those in larger, more financialized economies may exhibit greater residual variance than smaller, more stable ones. This violates the classical OLS assumption of homoskedasticity, where error variance is assumed constant across all observations.

To formally verify these concerns, we conduct the Wald Test for groupwise heteroskedasticity and serial correlation in panel data. This confirms the presence of these issues at conventional significance levels. We therefore report clustered standard errors at the country level, which relaxes both the homoskedasticity assumption and the assumption of no serial correlation within clusters simultaneously. This approach ensures that our analysis remains valid and our standard errors accurately reflect the true uncertainty in our coefficient estimates, without requiring us to alter the underlying fixed effects specification.

Furthermore, given the time-series dimension of our panel spanning 2010–2023, within-country serial correlation is also a concern. Observations within the same country across consecutive years are unlikely to be independent. For example, a positive shock to housing price growth in one year is likely to persist into the following year, inducing correlation in the error terms over time. This violates the assumption of independently distributed errors and can lead to underestimated standard errors, inflating t-statistics and overstating the precision of our estimates.

During the econometric model, we didn't transform our variables by applying log or with interaction term for this model. Because most of the variables we have are already in percentage unit, we felt logging was not necessary. In addition, we couldn't apply interaction term because there was no specific policy or comparison conducted based on certain period (such as Pre- and Post-COVID). However, if we were to compare the changes or implication of the policy, we could narrow the entities to the countries that implemented policy, and balanced year of pre and post policy.

RESULTS

Table 4a presents the results across six models — two pooled OLS models, two random effects model with and without time fixed effects, and two fixed effects models with and without time fixed effects. The coefficient on the old age dependency ratio (*oad_r*), our key explanatory variable, changes substantially in both sign and magnitude across specifications, reflecting the importance of controlling for unobserved country heterogeneity.

In the baseline pooled OLS models (1) and (2), the old age dependency ratio enters positively and statistically significantly, with coefficients of 0.707 and 0.448 respectively. This suggests that a one percentage point increase in the old age dependency ratio is associated with an increase of approximately 0.45 to 0.71 points in the real house price index, depending on the other explanatory variables considered in the model or not. However, these estimates are likely to be overestimated due to omitted country-level characteristics. Countries with older populations may simultaneously have higher incomes or stronger housing demand for structural reasons unrelated to aging itself. This overestimation in coefficients in pooled OLS is consistent with the broader panel data literature, which cautions against cross country and year comparisons that fail to account for unobserved heterogeneity.

The random effects models tell an instructive story about the role of time fixed effects. Model (3), estimated without year dummies, produces a positive and significant coefficient of 0.973 on *oad_r* with p-value significant at 5% level. However, once time fixed effects are introduced in Model (4), the coefficient flips to -0.469 and loses statistical significance. This reversal in sign between Models (3) and (4) reveals that time trends were confounding the RE estimate in Model (3), potentially inflated the coefficient upward, or overestimated. Without absorbing these common shocks through year dummies, the aging effect appears positive because older populations and rising housing prices happened to trend together over the sample period. The model fit, or within R^2 , increased to 0.4863, and 0.6879, respectively. It showed a noticeable increase in the model fit with the time dummies in the random effects model.

Before moving on to the fixed effects model, we ran Wald Test to test the heteroskedasticity of the explanatory variables. After considering all the explanatory variables and applying entity and year fixed effects, the test resulted chi-square of 2827.76 with p-value of 0.00 (*Table 5*), which allowed us to reject the null hypothesis that error variance is constant across all countries. This confirms that the presence of groupwise heteroskedasticity, which is expected given the potential structural differences across the 31 OECD economies in our sample; countries with larger and competitive housing markets are likely to exhibit different residual variance compared to smaller economies. To correct for this, as well as for within-country serial correlation, all subsequent specifications report clustered standard errors at the country level.

Once we move to the fixed effects specifications, the picture changes considerably compared to the models with random effects. Model (5), which includes only entity fixed effects, produces a positive but statistically insignificant coefficient of 0.955, suggesting that after removing time-invariant country differences, the within-country relationship between aging and housing prices becomes unclear. Most importantly, Model (6), which is our preferred model that incorporates both entity and time fixed effects, produces a negative association and statistically significant coefficient of -2.316, with the p-value less than 0.05. This implies that a one percentage point increase in the old age dependency ratio is associated with a decrease of approximately 2.316 points in the real house price index, after controlling for macroeconomic conditions and absorbing both country-level unobservable characteristics and time trends.

To determine whether random effects or fixed effects is the more appropriate specification, we conduct the Hausman test. We used both the fixed effects and random effects to have time dummies for time fixed effects. *Table 6* reports a test statistic of $\chi^2(9) = 78.23$ with a p-value of 0.0000, reflecting rejecting the null hypothesis that the country-specific effects are uncorrelated with the explanatory variables. This shows that the fixed effects model should be utilized throughout this research, confirming that RE estimates would be inconsistent for our dataset. In addition, this aligns with each model's characteristics that the random effects are suitable if the key variable of interest is time constant; in our research, our main explanatory variable – old age dependency ratio – varies across time, which makes sense to use fixed effects model more. Based on the Hausman Test's result, the next supplemental regression models will be mostly based on the fixed effects model as well.

Turning to the other explanatory variables, *core_cpi* is positive and statistically significant in Models (2) and (5), with a coefficient of 4.515 and 1.437 respectively, consistent with our earlier discussion that housing prices respond more than 1:1 proportionally to inflation: a coefficient well above 1 suggests housing serves as an inflation amplifier. The *avg_salary* variable switches from negative in OLS to strongly positive in both FE models, reflecting again the omitted variable bias in pooled OLS. Within countries, wage growth is positively associated with

housing price growth, as higher purchasing power drives demand. However, we need to be cautious that though the average salary level increases, the inflation rate could affect the real wage level. The *net_save* variable carries a negative sign consistently across all models, and is statistically significant in Models (2) and (4) at 5% significance level or better, suggesting that countries with higher net savings rates — measured as gross national savings minus depreciation as a percentage of GDP — tend to experience lower housing price growth, potentially reflecting that economies with stronger aggregate capital accumulation channel investment away from the housing sector toward more productive uses. Finally, the *unemp_rate* is negative and significant in Model (6), indicating that within-country increases in unemployment suppress housing price growth, consistent with a demand-side interpretation of housing markets.

From the Model (6), 2021 stands out with the largest and most statistically significant coefficient of 37.74, significant at the 1% level. This suggests that housing prices in 2021 were substantially higher than in the base year of 2010, holding other variables constant. Considering the period of 2020 and 2021 and *Plot 1*, this is consistent that housing boom occurred which had upward pressure in housing price.

To further examine the robustness of our main findings and explore the relationship between housing prices and the explanatory variables under alternative specifications, *Table 4b* presents three supplemental models in which certain variables are selectively excluded from the models from *Table 4a*. The models in this table are not intended to replace the main specifications, but to assess whether the key relationships hold when the model composition changes by excluding some other explanatory variables.

Model (1) from *Table 4b* replicates the full OLS specification from the main results but excludes *unemp_rate*, which was insignificant in *Table 4a*'s Model (2). The coefficient on *oad_r* remains positive and statistically significant at 0.447 with p-value less than 0.01, closely mirroring the earlier OLS estimate of 0.448, which reinforces that the overestimation in pooled OLS is consistent and not driven by any explanatory variable. Explanatory variable *core_cpi* still remained a large positive coefficient of 4.589 – a 0.074 decrease from the *Table 4a*'s Model (2) – suggesting that inflation is strongly associated with housing price growth. However, as mentioned earlier, this likely reflects omitted country-level bias rather than a causal relationship. Models (2) and (3) from *Table 4b* are both fixed effects models with entity and year fixed. Model (2) excludes *core_cpi* but retains *real_gdp*, while Model (3) excludes both *core_cpi* and *real_gdp*. The Model (2) was structured as the *Table 4a*'s Model (6)'s core CPI had the lowest t-statistics, meaning it has the least statistical significance. Across both specifications, the coefficient on *oad_r* remains negative and statistically significant at approximately -2.19 and -2.23 respectively, and both statistically significant at 10% level, remarkably close to the -2.316 estimated in our preferred main specification. This small changes across the models provide reassurance that the negative association between the old age dependency ratio and real housing prices is not driven by any particular choice of control variables, but it could be a robust finding. The *avg_salary* and *lt_ir* variables also maintain their signs and significance across both fixed effects supplemental models, consistent with what was observed in the main results. This sign reversal after applying fixed effects further reinforces the earlier interpretation that pooled OLS confounds between-country structural differences with true within-country level. The *unemp_rate* coefficient strengthens in its magnitude slightly when *real_gdp* is dropped in Model (3), becoming significant at the 1% level with a coefficient of -1.987, suggesting that unemployment and real GDP per capita capture partially overlapping element in housing

demand, and that unemployment's independent effect becomes stronger when real GDP per capita is excluded from the specification.

From the Model (3) from *Table 4b*, we got slightly higher coefficient in year 2021 – 39.48 – and this model resulted the highest time coefficient in 2022, which was 39.67. Both years were statistically significant at 1% level. These results remain consistent with the upward trend observed in *Plot 1*, and reflect the increasing housing price pressures that extended after the initial COVID-19 shock into 2022.

CONCLUSION

This paper examines the association between the old age dependency ratio and real housing prices across 31 OECD economies from 2010 to 2023, motivated by growing concerns over housing affordability and intergenerational wealth inequality, particularly in the context of South Korea's rapidly aging demographic structure. Employing an entity and year fixed effects model selected based on the result of Hausman test, we find a negative and statistically significant relationship between the old age dependency ratio and the real house price index. Specifically, a one percentage point increase in the old age dependency ratio is associated with a decrease of approximately 2.316 points in the real house price index, after controlling for long-term interest rates, unemployment, average salary, net savings rate, and real GDP. This finding is robust across supplemental specifications in which key control variables are selectively excluded, with the coefficient on the old age dependency ratio remaining stable at approximately -2.2 across alternative model compositions.

The broader implications of this finding are significant. As OECD economies continue to age, our results suggest that demographic structure change to older society may influence downward pressure on real housing prices over the long run. This has important consequences for older homeowners who rely on housing wealth as a primary store of retirement savings, as well as for policymakers tasked with maintaining housing market stability. At the same time, declining housing prices driven by aging demographics may offer some relief to younger generations who have been increasingly priced out of homeownership, which is the situation that is reflected in South Korea, where the average age of first-time buyers has risen by over 14 years since the early 1990s. Policymakers should therefore consider demographic trajectories and progress as a structural input into housing policy, rather than treating aging solely as a social welfare concern. Nevertheless, this study carries several limitations. First, the old age dependency ratio is a broad measure of demographic aging. While it captures the ratio of elderly to working-age population, it fails to distinguish between the housing behavior based on age levels. For example, those aged 65 to 74 versus those aged 75 and older, who may have meaningfully different patterns of housing asset divestment. Second, while the two-way fixed effects model controls for time-invariant country characteristics and common time trends, it cannot account for time-varying country-specific shocks, such as sudden change in national housing policy, zoning law changes, or demographic shocks like rapid immigration, that may have confounding elements in the estimated relationship. Third, our sample is restricted to OECD economies, which share broadly similar institutional and economic status. The findings may not generalize to developing countries or lower-income countries with different housing market structures and demographic dynamics. Fourth, while we document an association, our design does not reflect causal relationship in the strict sense. Compared to *Cho's* paper that examines the causal relationship between population aging and housing prices in South Korea, our research paper does not directly assess the causality. Finally, the elasticity of housing supply and demand are not

considered. As the slope of these lines reflect the elasticity, the shift in the supply and demand will rely on the elasticity which will determine the equilibrium housing price.

Several extensions on this research could follow from this work. First, to make the research more detailed, it could disaggregate the old age dependency ratio into age groups by separating those aged 65 to 74 from those aged 75 and older, which is also mentioned in the limitations, to better capture the stage of the life cycle at which housing divestment accelerates. Additionally, incorporating housing supply-side variables such as construction rates or zoning stringency indices would allow the model to distinguish between demand-driven and supply-driven components of housing price movements, providing a more complete picture of the demographic-housing nexus. Future research could also explore another model, such as difference-in-difference or instrumental variable model. This way, the extended research can observe whether the negative association between the old age dependency ratio and housing prices reflects a true causal relationship, rather than simple correlation, by more credibly isolating the effect of demographic aging from confounding factors. Finally, extending the analysis beyond OECD economies to include a broader set of countries would allow researchers to assess whether the findings generalize across different stages of economic development and demographic transition.

From a policy perspective, our findings carry meaningful implications for housing market policies and wealth distribution across OECD members. The negative association between the old age dependency ratio and real housing price indicates that aging society may bring downward pressure on housing markets over time. While this could offer affordability relief to young home seekers, policymakers should be cautious about treating declining housing prices. For older generations or homeowners who might rely on housing wealth as their retirement income or treat as one of their investments, a structural decline in housing prices might bring risks to their retirement plans and potentially increase fiscal pressure on public pension systems. At the same time, the negative relationship between the housing price index and unemployment infers the critical role of labor market conditions in sustaining housing demand. From a socioeconomic perspective, increasingly competitive job markets that delays young adults from entering employment might delay them to have independent homeownership, meaning the decrease in new housing demand might lead to depression in construction market. Policies that allow young adults not only to be employed but also allow them to actively participate in economy independently should be well considered, such as apprenticeship programs, housing subsidies for young adults, or targeted mortgage assistance for first-time home buyers.

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APPENDIX A: Tables and Plots

Plot 1: Scatterplot of the average old age dependency ratio and the real house price index, by year

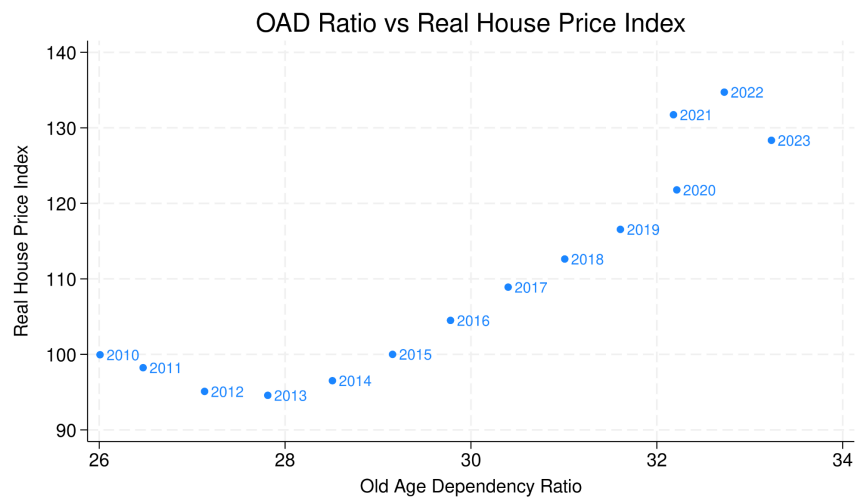


Table 1: Description of Variables

Variable Name	Description	Units
<u>house_ind</u>	Real House Price Index (measure the prices of residential properties over time)	Index (base year = 2015)
<u>oad_r</u>	Old Age Dependency Ratio (number of indivs aged 65 and over per working age(ages of 20-64))	Percent (%)
avg_salary	Average Annual Wages (PPP converted)	in 2024 USD
core_cpi	CPI, All items less food & energy, growth rate over one year	Growth Rate YoY in Percent (%)
lt_ir	Long-Term Interest Rates	Percent (%)
net_save	Net Saving (gross saving – depreciation)	Percent (%) of GDP
real_gdp	Real GDP per Capita (PPP converted)	in 2020 USD per person
unemp_rate	Unemployment Rate (15 years or older)	Percent (%)

Table 2: Descriptive Statistics of all variables

VARIABLES	(1) N	(2) mean	(3) sd	(4) min	(5) max
country_id	429	15.97	8.997	1	31
year	429	2,016	4.011	2,010	2,023
house_ind	429	110.0	18.49	69.70	181.3
oad_r	429	29.84	6.249	15.62	51.92
avg_salary	429	56.13	16.79	24.87	93.06
core_cpi	429	1.976	2.073	-3.096	13.76
lt_ir	429	2.395	2.329	-0.524	22.50
net_save	429	6.733	6.042	-12.77	25.66
real_gdp	429	51.47	19.46	23.08	128.3
unemp_rate	429	7.465	4.323	2.014	27.47

Table 3: Descriptive Statistics according to Pre / Post-COVID Years

	Pre-COVID (Mean)	Post-COVID (Mean)	Difference	p-value
house_ind	102.70	129.04	-26.35	0.000
avg_salary	55.23	58.47	-3.24	0.071
oad_r	28.79	32.58	-3.79	0.000
core_cpi	1.30	3.74	-2.44	0.000
lt_ir	2.62	1.82	0.79	0.000
net_save	6.61	7.04	-0.43	0.500
real_gdp	50.07	55.12	-5.06	0.021
unemp_rate	8.02	6.03	1.99	0.000
N	429			

Table 4: Main Regression Results

VARIABLES	(1) OLS	(2) OLS	(3) RE	(4) RE	(5) FE	(6) FE
oad_r	0.707*** (0.148)	0.448*** (0.157)	0.973** (0.481)	-0.469 (0.323)	0.955 (0.843)	-2.316** (1.117)
avg_salary		-0.235*** (0.0806)	-0.0308 (0.159)	0.0363 (0.138)	2.363*** (0.528)	1.411** (0.591)
core_cpi		4.515*** (0.475)	3.661*** (0.745)	1.970** (0.922)	1.437** (0.641)	1.165 (0.949)
lt_ir		-1.289* (0.679)	-1.005 (1.291)	1.372 (0.929)	-0.0930 (0.772)	1.277* (0.724)
net_save		-0.356** (0.154)	-0.538 (0.340)	-0.847*** (0.293)	-1.000* (0.518)	-0.631 (0.424)
real_gdp		0.200** (0.0800)	0.176 (0.119)	0.0565 (0.104)	0.594*** (0.199)	0.320 (0.195)
unemp_rate		-0.105 (0.239)	-1.058* (0.562)	-1.154* (0.598)	-1.143 (0.683)	-1.497** (0.609)
2011.year				-3.175** (1.593)		-2.208* (1.241)
				⋮	⋮	⋮
2023.year				19.18** (8.470)		27.22** (10.15)
Constant	88.91*** (4.343)	96.89*** (7.045)	80.18*** (16.69)	114.5*** (12.56)	-69.05** (33.49)	79.10 (51.52)
Observations	429	429	429	429	429	429
R-squared	0.057	0.337			0.634	0.739
Number of country_id			31	31	31	31
Entity FE			No	No	Yes	Yes
Year FE			No	Yes	No	Yes

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Year dummies are omitted between 2011 to 2023.

Table 4b: Supplemental Regression Results

VARIABLES	(1) OLS	(2) FE	(3) FE
oad_r	0.447*** (0.156)	-2.187* (1.113)	-2.227* (1.092)
avg_salary	-0.233*** (0.0802)	1.539** (0.575)	1.587** (0.613)
core_cpi	4.589*** (0.429)		
lt_ir	-1.368** (0.630)	1.449* (0.759)	1.522* (0.756)
net_save	-0.333** (0.142)	-0.700 (0.424)	-0.639 (0.403)
real_gdp	0.202** (0.0799)	0.333 (0.203)	
unemp_rate		-1.711** (0.665)	-1.987*** (0.621)
2011.year		-1.752 (1.229)	-1.622 (1.215)
		⋮	⋮
2023.year		30.93*** (10.27)	33.00*** (9.863)
Constant	95.83*** (6.889)	71.05 (52.59)	87.08* (48.92)
Observations	429	429	429
R-squared	0.337	0.735	0.731
Number of country_id		31	31
Entity FE		Yes	Yes
Year FE		Yes	Yes

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Year dummies are omitted between 2011 to 2023.

Table 5: Wald Test

Hausman Test	$\chi^2(31) = 2827.76$
	Prob > $\chi^2 = 0.0000$

Table 6: Hausman Test

Hausman Test	$\chi^2(9) = 78.23$
	Prob > $\chi^2 = 0.0000$

Appendix B: Stata Do

* Summary Statistics Ready

```
summarize country_id year house_ind oad_r avg_salary core_cpi lt_ir net_save real_gdp  
unemp_rate
```

```
outreg2 using "EC406 Summary Statistics.doc", sum(log) replace
```

* Scatterplot

```
collapse (mean) house_ind oad_r, by(year)
```

```
twoway scatter house_ind oad_r, mlabel(year) mlabposition(3) title("OAD Ratio vs Real House  
Price Index") ytitle("Real House Price Index") xtitle("Old Age Dependency Ratio")
```

```
restore
```

* Create Pre/Post COVID Dummy for Table 3

```
gen period = 0 if year < 2020
```

```
replace period = 1 if year >= 2020
```

```
estpost ttest house_ind avg_salary oad_r core_cpi lt_ir net_save real_gdp unemp_rate,  
by(period) unequal
```

```
esttab using EC406_Table3.rtf, cells("mu_2(label(Post-COVID) fmt(2)) mu_1(label(Pre-  
COVID) fmt(2)) b(label(Difference) fmt(2)) p(label(p-value) fmt(3))")
```

* Regression (OLS)

```
reg house_ind oad_r, robust
```

```
outreg2 using "EC406 Reg.doc", replace ctitle("OLS")
```

```
reg house_ind oad_r avg_salary core_cpi lt_ir net_save real_gdp unemp_rate, robust
```

```
outreg2 using "EC406 Reg.doc", append ctitle("OLS")
```

* xtset

```
xtset country_id year
```

*Random Effects

```
xtreg house_ind oad_r avg_salary core_cpi lt_ir net_save real_gdp unemp_rate, re vce(cluster  
country_id)
```

```
outreg2 using "EC406 Reg.doc", append ctitle("RE") addtext(Entity FE, No, Year FE, No)
```

```
xtreg house_ind oad_r avg_salary core_cpi lt_ir net_save real_gdp unemp_rate i.year, re  
vce(cluster country_id)
```

```
outreg2 using "EC406 Reg.doc", append ctitle("RE") addtext(Entity FE, No, Year FE, Yes)
```

* Wald Test

```
xtreg house_ind oad_r avg_salary core_cpi lt_ir net_save real_gdp unemp_rate i.year, fe  
vce(cluster country_id)
```

```
xttest3
```

* Fixed Effects

```
xtreg house_ind oad_r avg_salary core_cpi lt_ir net_save real_gdp unemp_rate, fe vce(cluster  
country_id)
```

```
outreg2 using "EC406 Reg.doc", append ctitle("FE") addtext(Entity FE, Yes, Year FE, No)
```

```

xtreg house_ind oad_r avg_salary core_cpi lt_ir net_save real_gdp unemp_rate i.year, fe
vce(cluster country_id)
outreg2 using "EC406 Reg.doc", append ctitle("FE") addtext(Entity FE, Yes, Year FE, Yes)
* Hausman Test
xtreg house_ind oad_r avg_salary core_cpi lt_ir net_save real_gdp unemp_rate i.year, re
estimates store re_model
xtreg house_ind oad_r avg_salary core_cpi lt_ir net_save real_gdp unemp_rate i.year, fe
estimates store fe_model
hausman fe_model re_model, sigmamore
* Additional Regressions
reg house_ind oad_r avg_salary core_cpi lt_ir net_save real_gdp, robust
outreg2 using "EC406 Reg2.doc", replace ctitle("OLS")
xtreg house_ind oad_r avg_salary lt_ir net_save real_gdp unemp_rate i.year, fe vce(cluster
country_id)
outreg2 using "EC406 Reg2.doc", append ctitle("FE") addtext(Entity FE, Yes, Year FE, Yes)
xtreg house_ind oad_r avg_salary lt_ir net_save unemp_rate i.year, fe vce(cluster country_id)
outreg2 using "EC406 Reg2.doc", append ctitle("FE") addtext(Entity FE, Yes, Year FE, Yes)

```